Solutions to JEE Advanced - 9 | JEE 2022 | Paper - 2

1.(BD) Normal reaction from ring $N = \frac{mV^2}{R}$

$$m\frac{dV}{dt} = -\mu N$$

$$\int_u^V \frac{dV}{V^2} = \int_0^t \frac{-\mu\, dt}{R} \Rightarrow \frac{1}{V} = \frac{1}{u} + \frac{\mu}{R}t \Rightarrow \frac{dt}{ds} = \frac{1}{u} + \frac{\mu}{R}t$$

$$\int_0^t \frac{dt}{\frac{1}{u} + \frac{\mu t}{R}} = \int_0^{\pi R} ds \Rightarrow \ln\left[1 + \frac{\mu u}{R}t\right] = \pi\mu$$

$$t = \frac{R}{\mu u}[e^{\pi\mu} - 1]$$

$$V = \frac{Ru}{R + \mu ut} = \frac{Ru}{R + R(e^{\pi\mu} - 1)} = ue^{-\pi\mu}$$

2.(AD) Pressure at base $= P = \rho gh \Rightarrow$ Force on base $= P \times \pi a^2 = \rho\pi gha^2$

Note that this force is exactly equal to the weight of cylindrical volume of water directly above the base [The cylinder of radius a and height h]

The weight of the rest of the water is supported by the curved surface.

Total weight of water in bucket

$$W = \rho\,[\text{Volume of frustum, } V]g.$$

$$V = \frac{\pi h}{3}[a^2 + ab + b^2]$$

Net force on curved surface $= W - \rho\pi gha^2$

Net force on bucket $= W$.

3.(BCD) The only possible point of collision is centre of square.

Now, for particle at D, possible time to reach the centre $= t_1 = \frac{(2n_1 - 1)d}{u_1}$; $\left(d = \frac{a}{\sqrt{2}}\right)$

And for particle at C, possible time to reach the centre $= t_2 = \frac{(2n_2 - 1)d}{u_2}$

For collision, $t_1 = t_2 \Rightarrow \frac{(2n_1 - 1)d}{u_1} = \frac{(2n_2 - 1)d}{u_2} \Rightarrow \frac{2n_1 - 1}{2n_2 - 1} = \frac{u_1}{u_2} = \text{odd}$

4.(ABC)

The number of waves encountered by the moving plane per unit time is given by

$$n = \frac{\text{distance travelled}}{\text{wavelength}} = \frac{c + v}{\lambda} = \frac{c}{\lambda}\left(1 + \frac{v}{c}\right) = f\left(1 + \frac{v}{c}\right)$$

The stationary observer meets the frequency f' of the incident wave and receives the reflected wave of frequency f'' emitted by the moving platform as

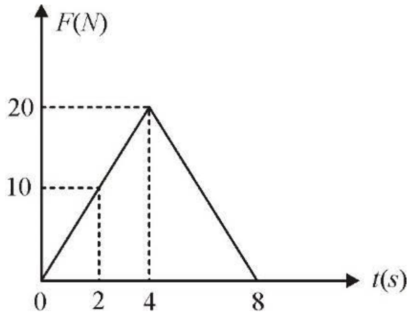
$$f'' = \frac{f'}{1-v/c} = \frac{f(1+v/c)}{1-v/c} = \frac{f(c+v)}{(c-v)}$$

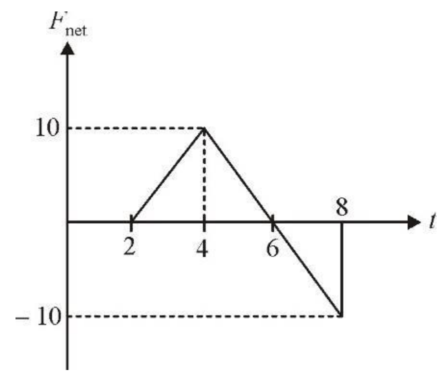
Wavelength, $\lambda'' = \frac{c}{f''} = \frac{c}{f}\left(\frac{c-v}{c+v}\right)$

Beat frequency = $f'' - f = \frac{f(1+v/c)}{(1-v/c)} - f = f\left(\frac{1+v/c}{1-v/c} - 1\right) = \frac{2vf}{c-v}$

5.(ABC) $\mu = 0.25$

$f_{\max} = 0.25 \times 4 \times 10 = 10N$





Block will start moving at 2s

$F_{net} = F_{applied} - friction$

Net impulse on the ball

From 2s to 8s

$= \frac{1}{2} \times 2 \times 10 = 10Ns$

= change in momentum

Velocity of the mass at t = 8s is 10/4 = 2.5 m/s

Block has maximum velocity at t = 6s

Net impulse on the ball from 2s to 6s = 20 Ns

Velocity of ball at (t = 6s) = 5 m/s. After t = 8s a constant retarding friction force of 10N acts and the block has momentum

= 10Ns at t = 8s.

Box will come to rest when impulse of 10Ns acts on it in the opposite direction of its motion

Box will come to rest 1s after t = 8s, i.e. at t = 9s.

6.(ABCD) $a = \frac{F}{m} = \frac{qE}{m} = \frac{q}{m}(\alpha - \beta x)$ …(i)

$a = 0 \; at \; x = \frac{\alpha}{\beta}$ i.e., force on the particle is zero at $x = \frac{\alpha}{\beta}$.

So, mean position of particle is at $x = \frac{\alpha}{\beta}$

Eq. (i) can be written as $v.\frac{dv}{dx} = \frac{q}{m}(\alpha - \beta x)$

$\therefore \int_0^v v\,dv = \frac{q}{m}\int_0^x (\alpha - \beta x)\,dx$ $\therefore$ $v = \sqrt{\frac{2qx}{m}\left(\alpha - \frac{\beta}{2}x\right)}$

$v = 0$ at $x = 0 \; and \; x = \frac{2\alpha}{\beta}$

So, the particle will oscillate between $x = 0$ to $x = \dfrac{2\alpha}{\beta}$

with mean position at $x = \dfrac{\alpha}{\beta}$.

Therefore, amplitude of particle is $\dfrac{\alpha}{\beta}$. Maximum acceleration of particle is at extreme positions (at $x = 0$ *or* $x = 2\alpha/\beta$) and $a_{\max} = q\alpha/m$ (from Eq. (1))

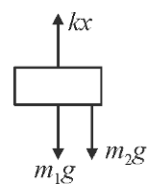
7.(ACD) $v_{rms} = \sqrt{\dfrac{3RT}{M}}, T = 300K, v_{rms} = 250\sqrt{3}\text{m/s}$

$\Rightarrow \quad M = 0.04$ kg/mol $= 40\ g$/mol

$$\rho = \frac{PM}{RT} = 1.6\ \text{kg/m}^3.$$

Kinetic energy of molecules $= \dfrac{nf}{2}RT = \dfrac{f}{2}PV = \dfrac{3}{2}PV = 30$ kJ

8.(BC) Draw force diagram of m_1 before string is cut.



$\Rightarrow \quad$ Elongation, $x = \dfrac{(m_1 + m_2)g}{k}$

When string is cut, the tension $(T = m_2 g)$ stops acting on m_1.

$\Rightarrow \quad$ Block m_1 accelerates up with $a = \dfrac{kx - m_1 g}{m_1} = \dfrac{m_2}{m_1}g$

At new equilibrium position, $kx' = m_1 g \quad \Rightarrow \quad x' = \dfrac{m_1 g}{k}$

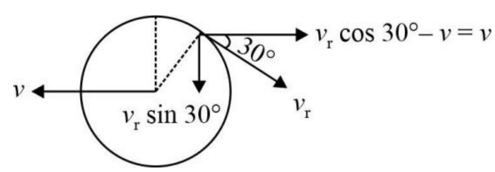
Amplitude of SHM $= x - x' = \dfrac{m_2 g}{k}$ Frequency of SHM $= \dfrac{1}{2\pi}\sqrt{\dfrac{k}{m_1}}$

After string is cut, block m_1 never goes below its initial position.

$\Rightarrow \quad$ Maximum elongation = initial elongation $= \dfrac{(m_1 + m_2)g}{k}$

9.(A) 10.(D)

Let V_r be the relative velocity of ball over the sphere. From conservation of linear momentum in horizontal direction.

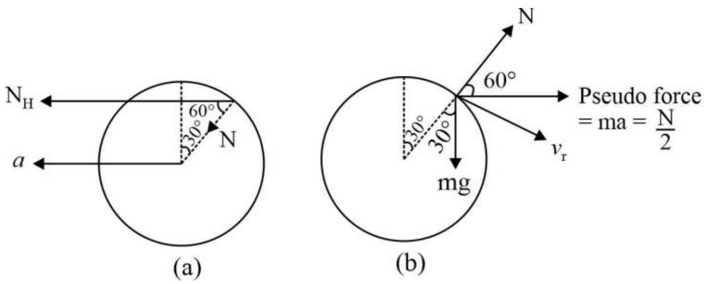


Momentum of sphere leftwards = momentum of ball rightwards or $mv = m(v_r \cos 30° - v) \quad \therefore \quad v_r = \dfrac{4v}{\sqrt{3}}$

Now, absolute velocity of ball at this instant will be $v_b = \sqrt{v_H^2 + v_v^2} = \sqrt{v^2 + (v_r \sin 30°)^2} = \sqrt{v^2 + \left(\dfrac{2v}{\sqrt{3}}\right)^2} = \sqrt{\dfrac{7}{3}}v$

Now, from conservation of mechanical energy, Decrease in PE of ball = increase in KE energy of ball and sphere $mgR(1 - \cos 30°)$

$$= \frac{1}{2}mv^2 + \frac{1}{2}m\left(\sqrt{\frac{7}{3}}v\right)^2 \text{ or } 10 \times 1\left(1 - \frac{\sqrt{3}}{2}\right) = \frac{5}{3}v^2 \text{ or } v \approx 0.9 m/s$$



Horizontal component of normal reaction is $N_H = N\cos 60° = \frac{N}{2}$

$\therefore$ Acceleration of sphere leftwards $a = \frac{N_H}{m} = \frac{N/2}{1} = \frac{N}{2}$

In figure (b) force diagram of ball with respect to sphere is shown.

$$mg\cos 30° - N - \frac{N}{2}\cos 60° = \frac{mv_r^2}{R} \quad \text{or} \quad 5\sqrt{3} - \frac{5N}{4} = \frac{16}{3}(0.9)^2 \quad \therefore\ N = 3.47N$$

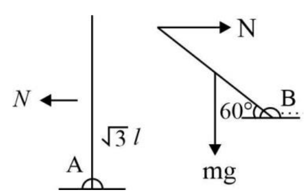
11.(D) $\frac{N_1}{V_1} = \frac{N_2}{V_2}$ (we can see from the table)

Hence, $N_2 = \frac{N_1}{V_1} \times V_2 = \frac{100}{120} \times 2400 = 2000$

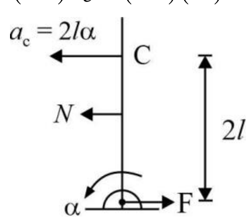
12.(B) $V_1 i_1 = V_2 i_2$ or $P_1 = P_2$ (we can see from the table)

13.(C) For rod 1: $\alpha = \frac{N(\sqrt{3}l)}{(2m)(16l^2)/3}$

or $N = \frac{32ml\alpha}{3\sqrt{3}}$



14.(A) $N - F = (2m)a_c = (2m)(2l)\alpha = 4ml\alpha$



$\therefore$ $F = N - 4ml\alpha = \left(\frac{32 - 12\sqrt{3}}{3\sqrt{3}}\right)ml\alpha$

15.(B) $i = qf = \left(\frac{2e}{3}\right)\left(\frac{v}{2\pi r}\right) = \frac{ev}{3\pi r}$

16.(A) $M = iA = \left(\frac{ev}{3\pi r}\right)(\pi r^2) = \frac{evr}{3}$

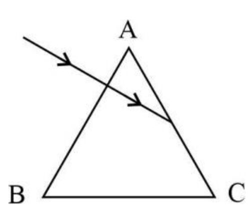
17.(A) In equilateral prism A = 60°

For normal incidence i_1 *and* r_1

Both are zero.

Further, $r_1 + r_2 = A$

$\therefore$ $r_2 = A = 60°$



Applying $\mu = \dfrac{\sin i_2}{\sin r_2}$

We have, $\sin i_2 = \mu \sin r_2 = \sqrt{3}\sin 60° = \dfrac{3}{2}$

but $\sin i_2 \not> 1$ $\therefore$ TIR will take place on the other face.

18.(C) **(P)** $U = \dfrac{1}{2}kT \Rightarrow ML^2T^{-2} = [k]K \Rightarrow [K] = ML^2T^{-2}K^{-1}$

(Q) $F = \eta A \dfrac{dv}{dx} \Rightarrow [\eta] = \dfrac{MLT^{-2}}{L^2LT^{-1}L^{-1}} = ML^{-1}T^{-1}$

(R) $E = hv \Rightarrow ML^2T^{-2} = [h]T^{-1} \Rightarrow [h] = ML^2T^{-1}$

(S) $\dfrac{dQ}{dt} = \dfrac{kA\Delta\theta}{\ell} \Rightarrow [K] = \dfrac{ML^2T^{-3}L}{L^2K} = MLT^{-3}K^{-1}$

19.(C) $\nu = n\left(\dfrac{c}{2\ell}\right) \Rightarrow n = \dfrac{2\ell\nu}{c}$

Number of nodes $= n+1$

Number of antinodes $= n$

20.(A) (P) 2, 3 (Q) 1, 3 (R) 1, 2, 3 (S) 4

CHEMISTRY

1.(B) Here, Pb^{2+} is the common ion in both, $Pb(NO_3)_2$ (strong electrolyte) and PbI_2 (weak electrolyte)

$$K_{sp}(PbI_2) = [Pb^{2+}][I^-]^2$$

$$1.4\times10^{-8} = [Pb^{2+}][I^-]^2 = (0.01)[I^-]^2$$

$$[I^-] = 1.18\times10^{-3}\,M$$

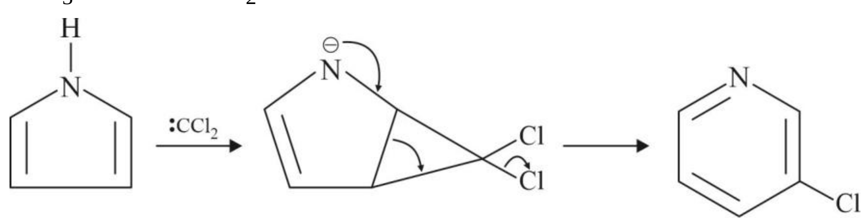
If S is the solubility of PbI_2, then $[I^-] = 2s$

$$PbI_2 \rightleftharpoons Pb^{2+} + 2I^-$$

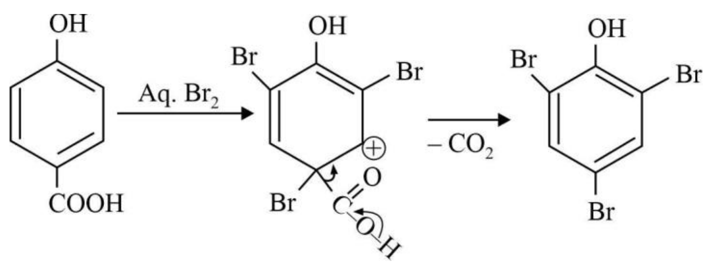
$$S \quad 2S = 1.18\times10^{-3}$$

$$S = 5.9\times10^{-4}\,M$$

2.(BD) $CHCl_3 + OH^- \rightarrow :CCl_2$



3.(B)



4.(B) With MeO^-/Δ Saytzeff product is major product. With Me_3CO^-/Δ

Hoff mann product is major product.

5.(ABD) (A) Fact

(B) $5O_3 + 2KOH \longrightarrow \underset{\text{(Orange coloured solid)}}{2KO_3} + 5O_2 + H_2O$

(D) $2O_3 \xrightarrow[\text{arc}]{\text{electric}} 3O_2$

6.(A) $^{68}_{32}G + {}^{0}_{-1}e \xrightarrow[\textit{capture}]{\textit{electron}} {}^{68}_{31}Y + X - rays + v$

7. (A) Self Reduction method

8.(AB) (A) As some ammonia is removed by 2nd reaction, some N_2H_4 would decompose to restore 1st equilibrium

(B) As $K_1 < K_2$, less ammonia will be formed than it is decomposed

(C) partial pressure of N_2 will increase but it can not be predicted with guarantee that it would be doubled.

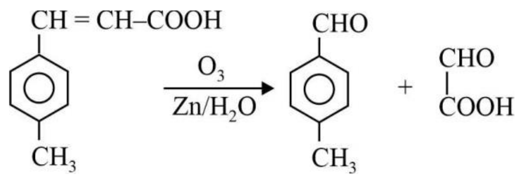
9.(A) Both Pb^{+2} and Cu^{+2} belongs to II group and precipitated as black metal sulphide Mn^{+2}, Ni^{+2} and Fe^{+3} are NOT in II group.

10.(D) Due to very low solubility $(ksp = 10^{-64})$ HgS is insoluble in yellow ammonium sulphide.

11.(A) Structure of X is

Which gives only one mononitrating product

12.(D)



13.(B) As $T_A = T_B$ thus $\Delta S_{A \to B}$

14.(A) A to C having constant volume therefore these process is isochoric

C to B having constant pressure therefore these process is isobaric.

15.(C) $\underset{(A)}{Hg_2(NO_3)_2} + 2NaCl \longrightarrow \underset{(B)\text{ white ppt}}{Hg_2Cl_2 \downarrow} + 2NaNO_3$

16.(B) $\underset{(A)}{Hg_2(NO_3)_2} + Na_2CrO_4 \longrightarrow \underset{\text{Red ppt}}{Hg_2CrO_4}$

17.(D) (P) $2NaBr + MnO_2 + 3H_2SO_4 \to 2NaHSO_4 + MnSO_4 + 2H_2O + Br_2$

(Q) $I_2 + 5Cl_2 + 6H_2O \to 2HIO_3 + 10HCl$

(R) $CaOCl_2 + 2CH_3COOH + 2KI \to (CH_3COO)_2Ca + 2KCl + I_2 + H_2O$

(S) $4NaCl + K_2Cr_2O_7 + 6H_2SO_4 \longrightarrow 2CrO_2Cl_2 + 4NaHSO_4 + 2KHSO_4 + 3H_2O$

18.(A) P-3; Q-4; R-1; S-2

19.(C) (P) $PhCOOH + PhNH_2 \longrightarrow PhCOO^-(aq) + PhNH_3^+(aq)$

As $PhCOOH$ is a weak acid, its conductivity is already less. On addition of weak base, acid-base reaction takes place and new ions are created. So conductivity increases.

(Q) $NaCl(aq) + AgNO_3(aq) \longrightarrow AgCl \downarrow (ppt) + NaNO_3(aq)$

As the only reaction taking place is precipitation of AgCl and in place of Ag^+ Na^+ is coming in the solution, conductivity remain nearly constant and then increases.

(R) $KOH + HCl \longrightarrow KCl(aq) + H_2O$

As H^+ is getting replaced by K^+ conductivity decreases and after end point, due to OH^-, it increases.

(S) $NaOH + CH_3COOH \longrightarrow CH_3COONa(aq) + H_2O$

$OH^-(aq)$ is getting replaced by CH_3COO^-, which has poorer conductivity. So conductivity decreases and then after the end point, due to common ion effect, no further creation of the ions take place. So, conductivity remain nearly same.

20.(B) (P) $\Delta G^0_{MnO_4^-/MnO_2} = \Delta G_1^0 + \Delta G_2^0$

$\Rightarrow$ $-3\times F\times E° = (-5\times 1.51\times F) + (2\times 1.23\times F)$

$\Rightarrow$ $-3E°F = -7.55F + 2.46F$

$\Rightarrow$ $-3E° = -5.09$

$\Rightarrow$ $E° = 1.69$

(Q) $\Delta G^0_{ClO_3^-/Cl^-} = \Delta G_1^0 + \Delta G_2^0$

$\Rightarrow$ $-6E°F = (-4\times 0.54F) + (-2\times 0.76\times F)$

$\Rightarrow$ $-6E° = -2.16 - 1.52$

$\Rightarrow$ $E° = 3.68/6$

$\Rightarrow$ $E° = 0.61$

(R) $\Delta G^0_{(Au^+/Au)} = \Delta G_1^0 + \Delta G_2^0$

$\Rightarrow$ $-1\times E°\times F = -1.5\times 3\times F + (1.4\times 2\times F)$

$\Rightarrow$ $-E° = -4.5 + 2.8$

$\Rightarrow$ $E° = 1.7$

(S) $\Delta G^0_{(Cr^{3+}/Cr^{2+})} = \Delta G_1^0 + \Delta G_2^0$

$\Rightarrow$ $-1\times F\times E° = -(-0.74\times 3\times F) + (-0.91\times 2\times F)$

$\Rightarrow$ $-E° = 2.22 - 1.82$

$\Rightarrow$ $-E° = 0.4$

$\Rightarrow$ $E° = -0.4$

MATHEMATICS

1.(BD) We have $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2\left(a+\frac{k}{n}\right)\left(b+\frac{k}{n}\right)}$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{1}{\left(a+\frac{k}{n}\right)\left(b+\frac{k}{n}\right)}$$

$$=\int_0^1\frac{1}{(a+x)(b+x)}dx=\frac{1}{a-b}\int_0^1\frac{(a+x)-(b+x)}{(a+x)(b+x)}dx$$

$$=\frac{1}{a-b}\int_0^1\left(\frac{1}{b+x}-\frac{1}{a+x}\right)dx$$

$$=\frac{1}{a-b}\left[\ln(b+x)-\ln(a+x)\right]_0^1=\frac{1}{a-b}\left[\ln\frac{(b+x)}{(a+x)}\right]_0^1$$

$$=\frac{1}{a-b}\ln\frac{(b+1)a}{(a+1)b}$$

Now if $a=b$ then $l=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{n}\frac{1}{\left(a+\frac{k}{n}\right)^2}=\int_0^1\frac{dx}{(a+x)^2}$

$$=\left[\frac{1}{a+x}\right]_1^0=\frac{1}{a}-\frac{1}{a+1}=\frac{1}{a(a+1)}$$

2.(AB) Let AB and CD be the line

$y=x$ …(i)

and $y=-x$ …(ii)

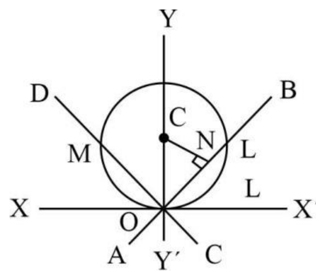
From the figure it is clear that by symmetry, one centre will be on the y-axis. Let the coordinates of the centre C be $(0,b)$.

As the circle passes through the origin, its radius will be b.

Draw CN perpendicular to AB.

Now, $ON=\left(\frac{1}{2}\right)OL=\left(\frac{1}{2}\right)a$

Length of the perpendicular from (0, b) on $y-x=0$ $=CN=\frac{b-0}{\sqrt{(1+1)}}=\frac{b}{\sqrt{2}}$



In triangle $CNO, CO^2=CN^2+ON^2$

or $b^2=\frac{b^2}{2}+\left(\frac{a}{2}\right)^2$

or $b=\pm\frac{a}{\sqrt{2}}$

The co-ordinates of the centre will be $\left(0,\pm\frac{a}{\sqrt{2}}\right)$ and radius $=\frac{a}{\sqrt{2}}$.

Hence, the equation of the circle is $(x-0)^2+\left(y\pm\frac{a}{\sqrt{2}}\right)^2=\left(\frac{a}{\sqrt{2}}\right)^2$

or $\quad x^2+y^2\pm\sqrt{2}ay=0$

Similarly, if the centre lies on x-axis the equation will be $x^2+y^2\pm\sqrt{2}ax=0.$

3.(AB) $\vec{r}.\vec{n}_1=q$ and $\vec{r}.\vec{n}_2=q,\ \vec{r}.\vec{n}_3=q$

Intersect in a line if $\left[\vec{n}_1\ \vec{n}_2\ \vec{n}_3\right]=0$. So,

$$\begin{vmatrix}1 & 1 & 1\\ 1 & 2a & 1\\ a & a^2 & 1\end{vmatrix}=0$$

or $\quad 2a-a^2-1+a+a^2-2a^2=0$

or $\quad 2a^2-3a+1=0$

or $\quad a=1/2,1$

4.(CD) Given that $\sin A+\sin B=\frac{c(a+b)}{c^2}=\frac{a+b}{c}=\frac{\sin A+\sin B}{\sin C}$

$\Rightarrow \quad \sin C=1$

$\Rightarrow \quad C=\pi/2$

$\Rightarrow \quad$ For $\quad A+B=\pi/2$

$\Rightarrow \quad B=\frac{\pi}{2}-A \quad \Rightarrow \quad \sin B=\cos A$

Thus, we have $\sin A+\cos A=\frac{a+b}{c}$

5.(ABCD)

We have $3+3i\in A\ but \notin B$

Hence $n(A\cap B)=0$

6.(BCD)

$4-x>0, 1+x>0\Rightarrow x\in(-1,4)$ …(i)

Let $|x-1|<1\Rightarrow x\in(0,2)$ …(ii)

The inequality implies

$\log_2(4-x)>\log_2(1+x)\Rightarrow 4-x>1+x\Rightarrow x<\frac{3}{2}$ ...(iii)

(i) – (iii) $\Rightarrow \quad x\in\left(0,\frac{3}{2}\right)$

Let $|x-1|>1\Rightarrow x\in(-\infty,0)\cup(2,\infty)$ …(iv)

The inequality implies

$\Rightarrow \quad \log_2(4-x)<\log_2(1+x)$

$\Rightarrow \quad 4-x<1+x\Rightarrow x>\frac{3}{2}$ …(v)

(i), (iv), (v) $\Rightarrow \quad x\in(2,4)$

Finally, we have $x\in\left(0,\frac{3}{2}\right)\cup(2,4)-\{1\}$

7.(AC) $A+adj\left(B^T\right)=\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}$ ...(i)

$A^T-adj(B)=\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$

$\Rightarrow \quad A-(adj\ B)^T=\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ ($\because$ take transpose)

$\Rightarrow \quad A-adj(B)^T=\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ ...(ii)

Add eqns. (i) and (ii)

$\Rightarrow \quad 2A=\begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}$

$\Rightarrow \quad A=\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}\Rightarrow |A|=0$

$\Rightarrow \quad A^2=\begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}\Rightarrow 2A$

Similarly, $A^n=2^{n-1}A$

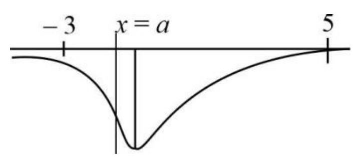
$\Rightarrow \quad 2adjB^T=\begin{bmatrix}2 & 0\\0 & 2\end{bmatrix}=2I$

$\Rightarrow \quad (adjB)^T=I$

$\Rightarrow \quad adj\ B=I$

$\Rightarrow \quad$ B = I

8.(ABCD) $f(x)=\dfrac{1}{2ax-x^2-5a^2}=\dfrac{1}{-4a^2-(x-a)^2}$



Clearly $f(x)$ is continuous $\forall x\in R$.

The graph is symmetrical about the line x = a.

If $a=1$ (mid point of $x=-3$ *and* $x=5$), greatest value is $f(5)=f(-3)$

If $a<1$, $f_{\max}==f(5)=\dfrac{1}{5\left(a^2-2a+5\right)}$

and if $a>1$, $f_{\max}=f(-3)=\dfrac{-1}{5a^2+6a+9}$

9.(B) $f(x)\le 0$ *and* $F'(x)=f(x)$

Now, $f(x)\ge cF(x)$

or $\quad F'(x)-cF(x)\ge 0$

or $\quad e^{-cx}F'(x)-ce^{-cx}F(x)\ge 0$

or $\quad \dfrac{d}{dx}\left(e^{-cx}F(x)\right)\ge 0$

Thus, $e^{-cx}F(x)$ is an increasing function

$\therefore \quad e^{-cx}F(x)\ge e^{-c(0)}F(0)$

or $\quad e^{-cx}F(x)\ge 0$

or $F(x) \geq 0$

or $f(x) \geq 0$ [as $f(x) \geq cF(x)$ and c is positive]

$\therefore$ $f(x) = 0$

Also, $\left(\dfrac{dg(x)}{dx}\right) < g(x) \forall x > 0$

or $e^{-x}\dfrac{d(g(x))}{dx} - e^{-x}g(x) < 0$

or $\dfrac{d}{dx}\left(e^{-x}g(x)\right) < 0$

Thus, $e^{-x}g(x) < e^{-(0)}g(0)$

or $g(x) < 0$ [as g(0) = 0]

Thus, $f(x)$ = g(x) has one solution, $x = 0$

10.(C) $\left|x^2 + x - 6\right| = f(x) + g(x)$ or $\left|x^2 + x - 6\right| = g(x)$

Thus, no solution exists.

11.(D) Centre of circumcircle is (10, 8) and radius $2\sqrt{2}$

So, circle $x^2 + y^2 - 20x - 16y + 156 = 0$

12.(D) $C_1C_2 > r_1 + r_2$

Since circles S_1 *and* S_3 do not intersect to each other, common chord does not exist.

13.(D) $\because$ $|z + iw| \leq |z| + |iw| = |z| + |i||w| \leq 2$

$\therefore$ $|z + iw| = 2 \Leftrightarrow |z| = |w| = 1$

14.(D) $|z + iw| = 2 \Rightarrow |z| = |iw|$

$\Rightarrow$ $w = e^{i\theta}$ and $z = e^{i(\theta + \pi/2)}$

$\Rightarrow$ $w = \cos\theta + i\sin\theta$ and $z = -\sin\theta + i\cos\theta$

$\Rightarrow$ (D) is correct

15.(C) $P(E) = \sum_{i=1}^{10} P\left(\dfrac{E}{A_i}\right)P(A_i) = \sum_{i=1}^{10} \dfrac{i}{11} \times \dfrac{1}{10} = \dfrac{1}{110}\sum i = \dfrac{1}{110} \times \dfrac{10 \times 11}{2} = \dfrac{1}{2}$

16.(D) $P\left(\dfrac{A_6}{E}\right) = \dfrac{P\left(\dfrac{E}{A_6}\right)P(A_6)}{P(E)} = \dfrac{\dfrac{6}{11} \times \dfrac{1}{10}}{\dfrac{1}{2}} = \dfrac{6}{55}$

17.(A) **(P)** $\cot x < 0 \Rightarrow -\cot x = \cot x + \operatorname{cosec} x$

$\Rightarrow$ $2\cos x + 1 = 0$ $\Rightarrow$ $x = \dfrac{2\pi}{3}, \dfrac{4\pi}{3}$

But $\cot\dfrac{4\pi}{3} > 0$

(Note: $\cot x \geq 0$ is not possible)

Hence, $x = \dfrac{2\pi}{3}$ only satisfies above equation.

(Q) The curves $y = n|\sin x|$ and $y = m|\cos x|$ intersect at 4 points in $[0, 2\pi]$.

(R) The given equation is valid if $\dfrac{9x}{10\pi}$ is an integer.

$x = 10\pi, \dfrac{100\pi}{9} \quad \Rightarrow \quad x = \dfrac{110\pi}{9}$

Hence, only one solution

(S) $\dfrac{\sqrt{3}-1}{2\sqrt{2}\sin x} + \dfrac{\sqrt{3}+1}{2\sqrt{2}\cos x} = 2$

$\sin\dfrac{\pi}{12}\cos x + \cos\dfrac{\pi}{12}\sin x = \sin 2x$

$\sin 2x = \sin\left(x + \dfrac{\pi}{12}\right)$

$\therefore \qquad 2x = x + \dfrac{\pi}{12}$ or $2x = \pi - \left(x + \dfrac{\pi}{12}\right)$

$x = \dfrac{\pi}{12}$ or $3x = \dfrac{11\pi}{12} \quad \therefore \quad x = \dfrac{\pi}{12}$ or $\dfrac{11\pi}{36}$

Therefore, k is 12

18.(B) **(P)** We get common normal perpendicular to $y = x$.

So, slope $= -1 \Rightarrow x + y = 3a$

(Q) Tangent to the parabola, $y = mx + a/m$ passes through the point P(h,k).

$\Rightarrow \qquad m^2h - mk + a = 0$

It its roots are m_1 *and* m_2, then $m_1m_2 = +1$, locus is $x = a$.

(R) The tangents are $y = m(x+a) + a/m$ …(i)

And $y = -\dfrac{1}{m}(x+2a) - 2a,$ …(ii)

Subtracting from eqns. (i) to (ii), we get $x + 3a = 0$.

(S) If chord joining t_1 *and* t_2 subtends $90°$ at vertex, then $t_1t_2 = -4$. point of intersection of tangents is $\left[-at_1t_2, -a(t_1+t_2)\right]$ so the locus is $x = 4a$.

19.(A) **(P)** Take 2, 3, 5 outside.

(Q) $(\vec{a}+\vec{b}).\{(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})\} = (\vec{a}+\vec{b})(\vec{b}\times\vec{c}+\vec{b}\times\vec{a}+\vec{c}\times\vec{a}) = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

(R) $(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a}) = \left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{c} \quad (\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b}) = \left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{a} \quad (\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c}) = \left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{b}$

(S) $\left[\vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\ \ \vec{a}\times\vec{b}\right] = (\vec{b}\times\vec{c}).\{(\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b})\} = \left[\vec{a}\ \vec{b}\ \vec{c}\right]^2$

20.(C) **(P)** The given line is $x = 4y+5,\ z = 3y-6$

or $\qquad \dfrac{x-5}{4} = \dfrac{z+6}{3} = y$

or $\qquad \dfrac{x-5}{4} = \dfrac{y}{1} = \dfrac{z+6}{3} = \lambda$ (say)

Any point on the line is of the form $(4\lambda+5, \lambda, 3\lambda-6)$.

The distance between $(4\lambda+5, \lambda, 3\lambda-6)$ and $(5,3,-6)$ is 3 units (given). Therefore

$(4\lambda+5-5)^2 + (\lambda-3)^2 + (3\lambda-6+6)^2 = 9$

or $\qquad 16\lambda^2 + \lambda^2 + 9 - 6\lambda + 9\lambda^2 = 9$

or $\qquad 26\lambda^2 - 6\lambda = 0$

or $\qquad \lambda = 0, 3/13$

The point is (5, 0, – 6)

(Q) The equation of the plane containing the lines $\frac{x-2}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and parallel to $\hat{i} + 4j + 7k$ is

$$\begin{vmatrix} x-2 & y+3 & z+5 \\ 1 & 4 & 7 \\ 3 & 5 & 7 \end{vmatrix} = 0 \text{ or } x - 2y + z - 3 = 0$$

Point $(-1,-2,0)$ lies on this plane.

(R) The line passing through points $A(2,-3,-1)$ and $B(8,-1,2)$ is $\frac{x-2}{8-2} = \frac{y+3}{-1+3} = \frac{z+1}{2+1}$

Or $\qquad \frac{x-2}{6} = \frac{y+3}{2} = \frac{z+1}{3} = \lambda$

Any point on this line is of the form $P(6\lambda+2, 2\lambda-3, 3\lambda-1)$, whose distance from point $A(2,-3,-1)$ is 14 units. Therefore, $PA = 14$

or $\qquad PA^2 = (14)^2$

$(6\lambda)^2 + (2\lambda)^2 + (3\lambda)^2 = 196$ or $\quad 49\lambda^2 = 196$ or $\quad \lambda^2 = 4$ or $\qquad \lambda = \pm 2$

Therefore, the required points are (14, 1, 5) and $(-10,-7-7)$. The point farther from the origin is $(14,1,5)$

(S) Any point on line AB, $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4} = \lambda$

Is $M(2\lambda, 3\lambda+2, 4\lambda+3)$. Therefore, the direction rations of PM are $2\lambda-3, 3\lambda+3$ and $4\lambda-8$.

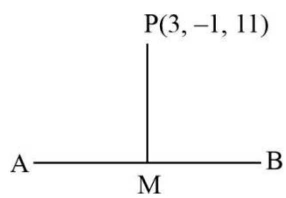
But $\qquad PM \perp AB$

$\therefore \qquad 2(2\lambda-3)+3(3\lambda+3)+4(4\lambda-8)=0$

$4\lambda-6+9\lambda+9+16\lambda-32=0$

$29\lambda-29=0$

$\lambda=1$



Therefore, foot of the perpendicular is $M(2, 5, 7)$.